

Calculating the Diffuse and Ambient Colors

Let's do some calculation involving diffuse and ambient lights. To simulate how a tiny part (face) of an object would look like to our eye due to ambient and diffuse lighting, consider its normal vector $\mathbf{N}$ and let's say a light ray is coming from direction $\mathbf{D}$ strikes this object. Then the diffuse colour of the light and observed colour of the small fragment will be given by:

vector $\mathbf{I}_{\text{ambient color}} = \text{ambient strength} * \text{lightcolor}$
vector $\mathbf{I}_{\text{diffusecolor}} = \max(\mathbf{N} \cdot \mathbf{D}, 0) * \text{lightcolor}$

vector $\mathbf{I}_{\text{result}} = (\text{ambientcolor} + \text{diffusecolor}) * \text{fragmentcolor}$

The resulting color vector is the colour perceived by our eye. (Note that ambient strength is a constant which describes the amount of ambient light.) Okay, so this was fairly simple, let's see how specular colour is calculated. Note that the last product is not a scalar or vector product, but it is the element-wise product of the two colour vectors

Calculating the Specular Color

The specular colour of a fragment as observed by the eye depends on the view direction relative to the normal of the fragment (Note that by fragment, we are referring to a tiny face on an object. Think of an object to be composed of many tiny faces). If the view direction is $\mathbf{V}$ (view position - fragment position), the normal of the fragment is $\mathbf{N}$ and the light ray

Hitting the fragment has direction $\mathbf{D}$, and the reflected ray from the the fragment has direction $\mathbf{D}'$, then we calculate the specular component of the objectcolor as:

vector $\mathbf{D}' = \text{reflect}(\mathbf{D}, \mathbf{N})$

vector $\mathbf{I}_{\text{specularcolor}} = \text{specularstrength} * [\max(\mathbf{V} \cdot \mathbf{D}', 0)]^{32} * \text{lightcolor}$

vector $\mathbf{I}_{\text{result}} = (\text{ambientcolor} + \text{diffusecolor} + \text{specularcolor}) * \text{lightcolor}$

Where specularstrength is a constant which depends on the object's material, this calculation was just done to show you a glimpse of how light interacts with objects so you can better model the lights and materials of the objects. Also, this calculation was done in the context of a very simple model (Phong), Blender has more complex implementations than these.

Types of Lights

Now, let's learn something about lights. Without knowing the different kinds of lights you can use in Blender, any knowledge about light interaction is